Day 2: Photosynthesis and Limiting Factors

Day 02 · From Lectures/photosynthesis-lecture.pdf · 2026-11-07

Overview of Photosynthesis

Photosynthesis converts light energy into chemical energy stored in glucose. It takes place in the chloroplasts of plant and algal cells.

Overall Equation

The overall equation is: 6 CO2 + 6 H2O + light energy -> C6H12O6 + 6 O2.

Light-Dependent Reactions

The light-dependent reactions occur in the thylakoid membranes of chloroplasts. Chlorophyll absorbs mainly blue and red light. Water is split via photolysis, which releases oxygen as a by-product. Energy from light is captured and stored as ATP and NADPH.

Calvin Cycle (Light-Independent Reactions)

The Calvin cycle takes place in the stroma. The enzyme RuBisCO fixes carbon dioxide onto RuBP (ribulose bisphosphate). ATP and NADPH supplied by the light-dependent reactions are then used to synthesize G3P, which is subsequently used to build glucose.

Limiting Factors

The rate of photosynthesis is limited by three primary factors: light intensity, carbon dioxide concentration, and temperature.